

Coordinate-sum invariant

Initial data
one singleton seed x
with $x_1 + x_2 = 1$

Closure under rule 1 and
integer linear combinations
adds only edge relations of
total label $(0, 0)$.

Hence every derived relation
has total label kx
for some $k \in \mathbb{Z}$.

Rule 2 needs a singleton
anti-diagonal vector
 $(a, -a)$ with $a \neq 0$.

Its total coordinate sum is 0.

If $kx = (a, -a)$ then $k(x_1 + x_2) = 0$. Since $x_1 + x_2 = 1$, we obtain $k = 0$, hence $(a, -a) = (0, 0)$, contradicting rule 2. **No new vertex can ever enter T .**